Advanced Quality Methodologies

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University
setu.ie

Advanced Quality Methodologies (A07601)

Short Title:	Advanced Quality Methodologies		
Faculty	Science and Computing		
Department	Science		
Cluster	Quality, Regulatory and Compliance		
Author	HHUGHES		
Credits	5	**Level:**	Advanced

Description of Module / Aims

This module covers aspects of quality management, including advanced quality systems such as Lean Manufacturing, Six Sigma and Risk Management, instrument validation and calibration management systems. It also looks at leadership, Innovation Management and Process Planning. It prepares the student for work in the pharmaceutical, healthcare or food industries, which operate under strict quality management regulations.

Programmes

SCIE-0026	BSc (Hons) in Applied Biology with Quality Management (WD_SQMBI_B)	4 / 2 / M
SCIE-0026	BSc (Hons) in Applied Chemistry with Quality Management (WD_SQMCH_B)	4 / 2 / M
SCIE-0026	BSc (Hons) in Food Science and Innovation (WD_SFSIN_B)	4 / 8 / M
SCIE-0026	BSc (Hons) in Food Science and Innovation (WD_SFDSC_B)	4 / 2 / M
SCIE-0026	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B)	4 / 8 / M
SCIE-0026	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	4 / 8 / M
	Higher Diploma in Science in Agi-Food ICT Systems (WD_18AFICT_G)	1 / 3 / M

Indicative Content

- Leadership, Change, Innovation, Strategic Management, Competitive advantage, Planning incl. Process Analysis, Affinity Diagrams
- TQM Gurus: Deming, Crosby and Juran
- Benchmarking, Quality Function Deployment
- Instrument validation, calibration management systems and traceability
- TQM: Lean Manufacturing, Lean thinking, tools such as Kaizen, TRIZ etc, Six Sigma
- Risk Management: Risk based thinking, Business Process reengineering
- Mini project involving a critical review of peer reviewed journal articles on a related topic

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Select appropriate leadership, planning and innovation management skills in an industrial environment.
2. Assess advanced quality management systems and their suitability in industry.
3. Critique current research in quality management.
4. Plan an instrument validation and calibration management system.

Learning and Teaching Methods

- Lectures.

- Independent research into related topics.
- Set examples.
- Mini project.
- Discussion.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	60%	1,2,4
Continuous Assessment	40%	
Assignment	40%	3

Assessment Criteria

<40%: No understanding of the basic concept of quality management. Unable to carry out critical thinking or problem solving techniques.

40%-49%: Is able to understand the quality management philosophies. May have some difficulty with critical thinking and problem solving.

50%-59%: All the above in addition to evaluation of quality systems and standards. Correctly make suggestions for correct instrument use in an industrial environment.

60%-69%: In addition to the above, good critical thinking and decision making abilities. Shows a high level of competence and efficiency in the quality management techniques.

70%-100%: All previous to excellent level. Be able to describe and discuss quality management in a production environment and show initiative.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Tutorial	12	
Independent Learning	87	

Essential Material

- "bambooweb." Total Quality Management. http://www.bambooweb.com
- "www.nsai.ie." http://www.iso.co.uk
- Mann, D. _Creating a Lean Culture: Tools to sustain lean conversions_. 3rd ed. Kentucky, USA: CRC Press, 2014.
- Miller, J.M. and J.B. Crowther. _Analytical Chemistry in a GMP Environment_. 1st ed. Ireland: John Wiley & Sons, 2000.
- Voehl, F., H.J. Harrington, C. Mignosa and R. Charron. _The Lean Six Sigma Black Belt Handbook_. 1st ed. Kentucky, USA: CRC Press, 2013.

Supplementary Material

- "European foundation of quality management." EFQM. http://www.efqm.org
- "Food and Drug Administration." http://www.fda.gov
- Boonstra, J.J. _Cultural Change & Leadership in organizations: A practical guide to successful organizational change_. 1st Edition. New Jersey, USA: Wiley Blackwell, 2013.
- Gryna, F.M., R.C.H. Chua and J.A. Defoe. _Juran's Quality Management & Analysis_. 6th ed. Ohio, USA: McGraw-Hill, 2014.
- Pyzdek, T. and P. Keller. _The Handbook for Quality Management: A complete guide to operational excellence_. 2nd ed. Ohio, USA: McGraw-Hill, 2012.